



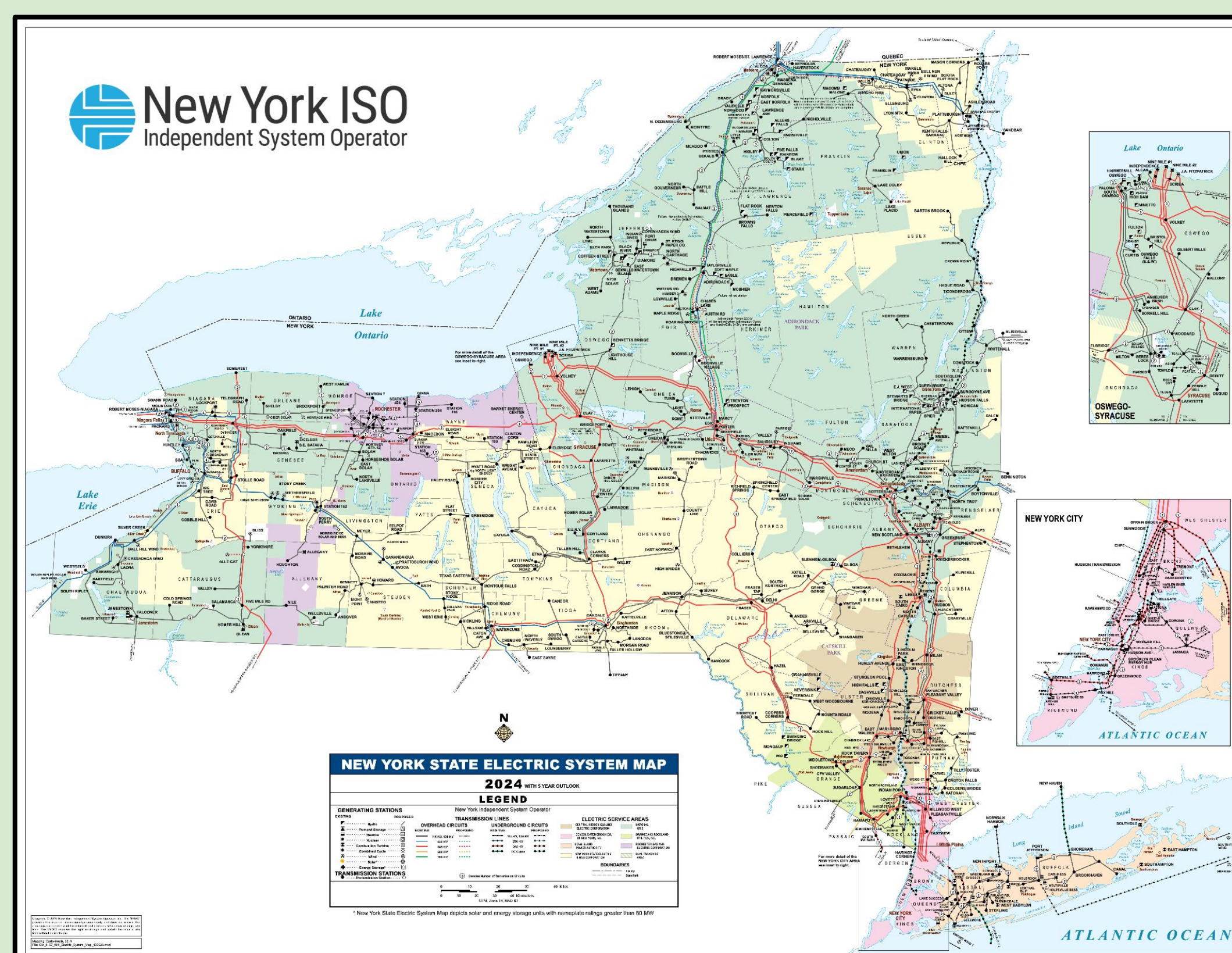
Dynamic Modeling of Northern New York Renewable Energy Sources

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Problem Statement

As of 2023, New York State has initiated a plan to achieve a power grid that is 70 percent renewable by 2030. Northern New York's significant hydroelectric and wind generation has been essential towards meeting these goals. Unfortunately, these renewable sources supply the New York power grid with instabilities, as perfect conditions cannot be secured for their operation. Additionally, due to the relatively small load in Upstate New York there is a large export of power to southern New York.



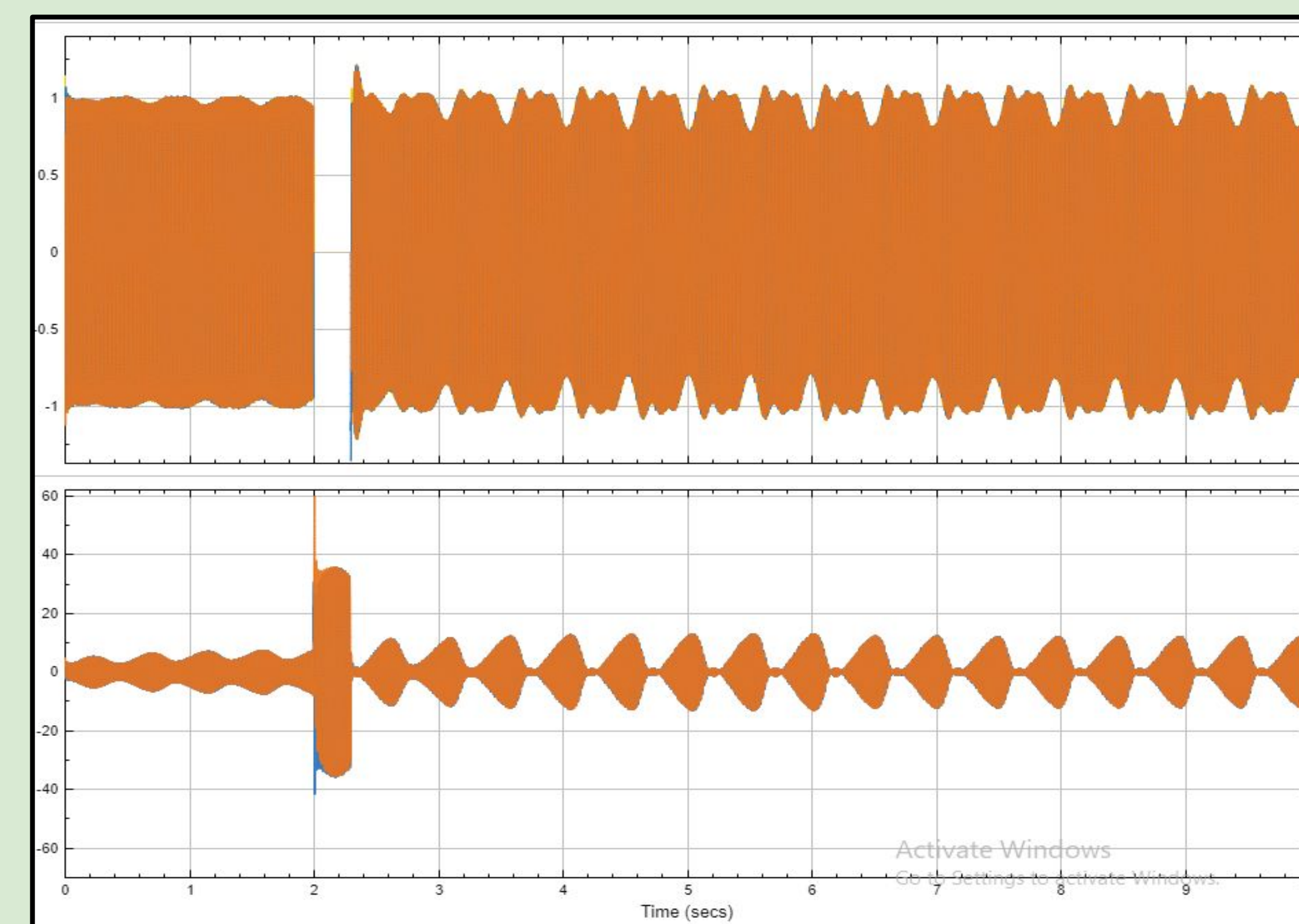
2024 New York State Electric System Map

Objectives

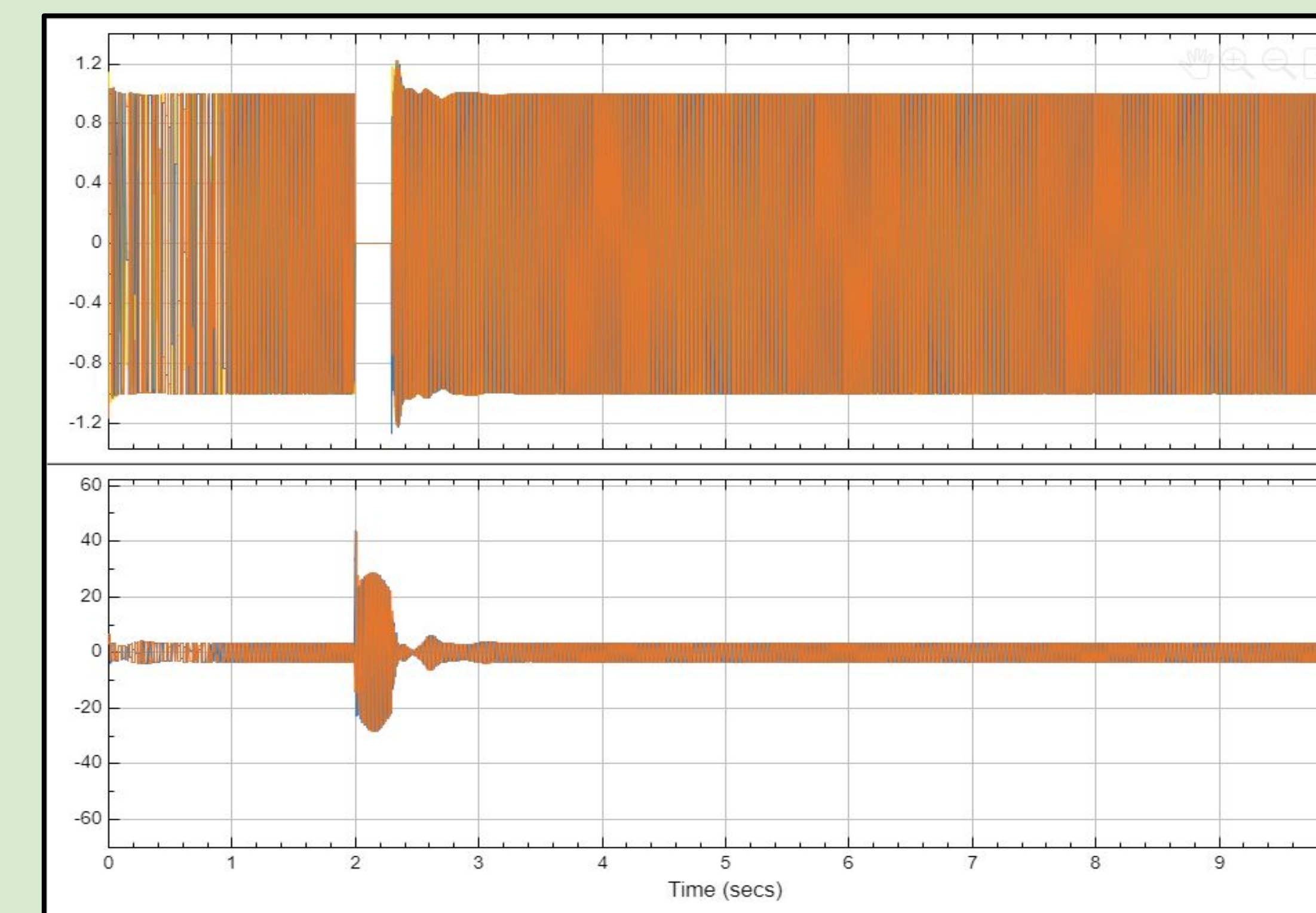
A dynamic model of the Northern New York power grid was built using MATLAB Simulink to investigate the potential for control interactions between these generation sources in the North Country and the downstate power grid. Consideration was also made for the high-voltage direct current (HVDC) tie with the power grid in Québec.

Fault testing of the dynamic model was conducted to determine the response of the renewable grid to undesirable conditions. This included: different types of faults, varying fault duration and simulated outages. For the results provided the fault duration was 0.3 seconds.

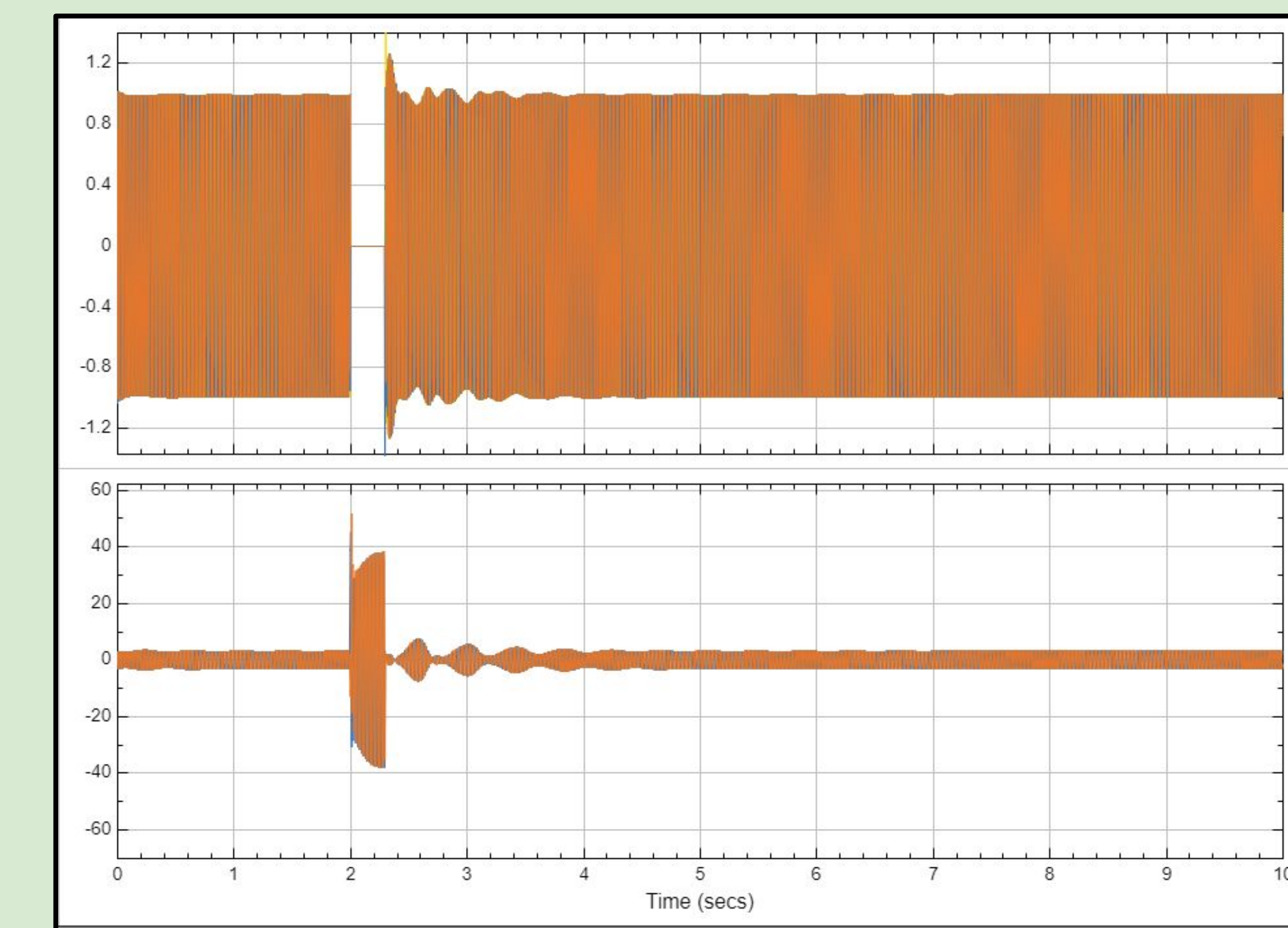
Stability Results for a 3-Phase Fault at the Massena Bus



Simulation with no 765 kV transmission lines



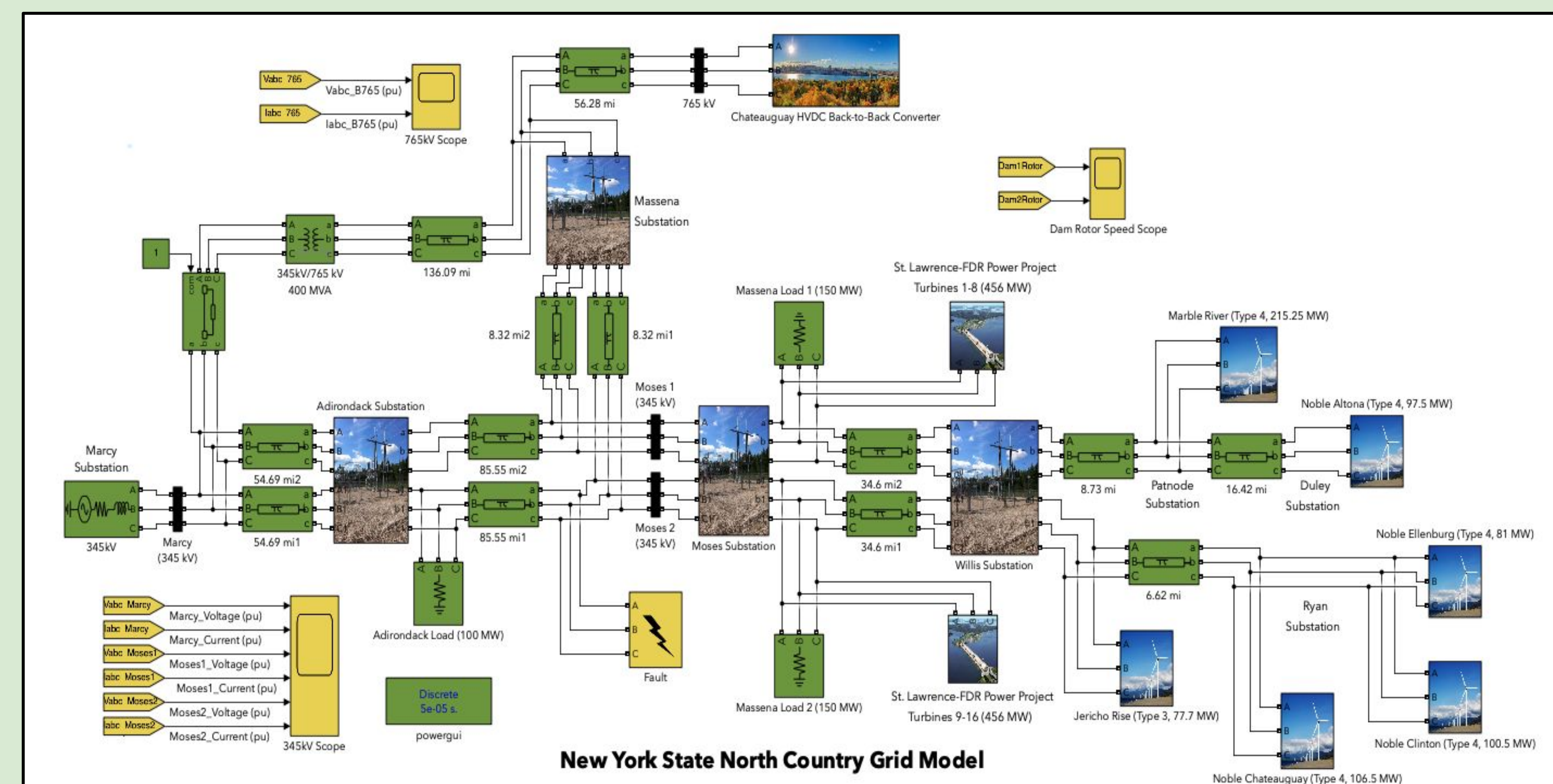
Simulation with no Wind Farms



Simulation with no 765kV line and Wind Farms

MATLAB Model

The most critical components of the system are 6 wind farms totaling 678.45 MW located near Chateaugay, NY and the St. Lawrence-FDR Power Project (912MW) hydro dam in Massena, NY. The New York State Power Authority (NYPA) is currently in the process of upgrading the transmission infrastructure from 230 kV to 345 kV to accommodate the addition of new renewable sources and a heavier downstate load, so the model is based around 345 kV transmission lines. The model includes Type III and IV wind farm subsystems designed by Canadian utility Hydro-Québec, a hydro power plant governor model with an DC excitation system, 345 kV and 765 kV transmission lines, the HVDC connection with Québec and circuit breakers for the control of the distribution system.



Conclusion

The most relevant conclusions to be drawn from the simulations performed are: the 765 kV transmission network and the HVDC tie with Hydro-Québec is essential for the NY state power grid, and that the electronics of the Wind Farms in the North Country do not have a strong effect on the stability of the grid. When the 765 kV transmission line is taken out the most effective solution is to also take the synchronous wind farms offline. Otherwise, the oscillations from the wind farm will be too strong for the hydro plant to damp on its own. With only the Moses Power Dam the grid stabilizes and provides downstate with a consistent source of power. The total power delivered to Marcy, 577.35 MW, is stabilized with the fault with the most efficiency by this hydro power plant. The most essential part of the Northern NY power grid is the hydro power produced in Massena, emphasizing the importance of providing funding for more hydro power in the future.

Acknowledgements

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References

